Matched-Frequency Synthetic Relations: Structured Affine Relation vs Random Memorized Relation

1 Question

We test whether two relations with identical frequency can be learned by different mechanisms. Relation R_0 is supported by latent attribute structure. Relation R_1 is a random permutation over the same subjects and objects. Both relations have the same number of distinct trained triples and the same repetition count.

2 Configuration

Quantity	Value
Subjects / objects	192 / 192
Attributes	$f \in \{0, \ldots, 31\}$, $g \in \{0, \ldots, 31\}$
Structured shift	$(\Delta_f, \Delta_g) = (3, 5)$
Relations	R_0 structured, R_1 random
Trained triples per relation	$k = 128$
Held-out subjects	64
Repeats per trained triple	128 for R_0, 128 for R_1
Attribute sentence repeats	4
Model	1-layer causal transformer
Hidden dimension / heads	64 / 4
Context length	8
Embedding tying	false
Learning rate	10^{-3}
Object loss weight	4.0

Table 1: Final matched-frequency configuration.

The run is configured in `experiments/configs/runs/synthetic_mixed_relations_category_k128_r128.tom` Training is ordinary end-to-end gradient descent. No components are frozen and no targeted component updates are used.

3 Data Generation

Each subject token a_i is assigned two latent attributes:

$$a_i \mapsto (f_i, g_i).$$

The structured target is a job-category object determined by the subject's f-attribute:

$$R_0(a_i) = b_{f_i}.$$

Thus many subjects can share the same structured target, analogous to many applicants with the same skill bucket matching the same job category.

For the category objects b_j, $j < 32$, object attributes are shifted:

$$b_j \mapsto (j + 3 \bmod 32,\ 0).$$

Remaining object tokens are distractor objects used by the random relation.

The model observes attribute sentences for all subjects and objects:

$$x\ a_i\ f_i\ g_i\ x,$$

$$x\ b_i\ f_i'\ g_i'\ x.$$

The structured relation is therefore supported by the latent skill/category attribute f_i.

The random relation is

$$R_1(a_i) = b_{\pi(i)},$$

where π is a seed-dependent random permutation. This relation is not supported by the attributes.

The relation training sentences are

$$x\ a_i\ r_0\ b_{f_i}\ x,$$

$$x\ a_i\ r_1\ b_{\pi(i)}\ x.$$

A seed-dependent shuffle selects 128 subjects for relation training and 64 subjects for held-out evaluation. Held-out subjects still appear in attribute sentences, but never appear in relation sentences.

4 Affine Test

Affine structure is evaluated with ordinary least-squares linear regression. For each relation, we fit

$$\hat{o} = Ws + b$$

from subject token embeddings s to the target object token embeddings o, using the 128 trained relation subjects. We then evaluate on the 64 held-out subjects.

The main metric is affine nearest-neighbor accuracy: the predicted vector $\hat{o}$ is compared against all object embeddings, and the prediction is correct if the nearest object is the held-out target.

5 Main Result

6 Replication

7 Interpretation

The frequency structure is matched exactly: both relations have 128 distinct trained triples and each trained triple is repeated 128 times. Both relations reach perfect top-1 recall on trained triples. R_0 also generalizes behaviorally to held-out subjects, while R_1 does not. Ordinary linear regression recovers the held-out structured mapping for R_0, while it fails for the random mapping R_1.

Thus, under matched frequency, the attribute-supported relation is represented with substantial affine structure, whereas the random relation is memorized without affine generalization.

Relation	Train top-1	Held-out top-1	Held-out top-10	OLS affine NN	OLS affine cosine
R_0 structured	1.000	0.875	0.953	0.969	0.910
R_1 random	1.000	0.000	0.000	0.016	0.214

Table 2: Seed 19. Both relations are learned on trained triples; only R_0 supports linear-regression affine generalization.

Seed	R_0 train	R_0 held-out	R_0 affine NN	R_1 train	R_1 held-out	R_1 affine NN
31	1.000	0.641	0.906	1.000	0.000	0.000
47	1.000	0.859	0.812	1.000	0.000	0.000
59	1.000	0.797	0.938	1.000	0.000	0.000

Table 3: Replication seeds preserve the same pattern.

8 Reproduction

```
cd experiments
env UV_CACHE_DIR=/tmp/uv-cache MPLCONFIGDIR=/tmp/mpl-cache \
  uv run factual-recall synthetic-mixed-relations \
  --config configs/runs/synthetic_mixed_relations_category_k128_r128.toml \
  --device cpu --quiet
```